CSE 318 — Online 2

Section B2

Your Greedy heuristic always starts by placing the endpoints of **one specific edge** (the heaviest edge) into X and Y. The rest of the construction follows from that choice. But the final result depends heavily on which edge you start with, and a different starting edge might give a better result.

Your task: Run the greedy heuristic multiple times, each time starting from a **different edge**:

1. Sort all edges by weight in decreasing order.
2. Take the top T edges. For each one:
   (a) Place its two endpoints into X and Y as the seed.
   (b) Run the greedy construction for the remaining vertices (same as your original greedy).
   (c) Record the resulting cut value.
3. Return the solution with the best cut value among all T runs.

This is called **Multi-Start Greedy**. It is deterministic (no randomness) and costs T times as much as a single greedy run.

What to run and report:

- Run Multi-Start Greedy with $T = 5, 10, 20$ on at least 6 benchmark graphs.
- Run your original single-seed greedy on the same graphs.
- Report the cut value before local search and after local search, for each.